LILaC: Late Interacting in Layered Component Graph for Open-domain Multimodal Multihop Retrieval

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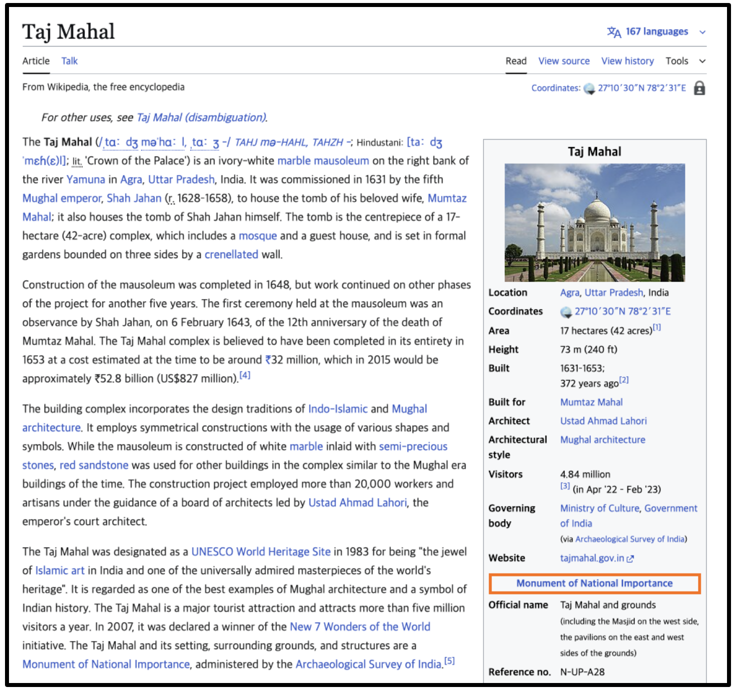
Multimodal Documents Are Everywhere!

- Real-world documents are often multimodal.
  - *e.g.,* webpages, PDFs, presentation slides, and way more!
- Accurate question-answering involves correctly resolving the relationships between the components.
  - *Component:* a multimodal element (i.e., text/table/image) which comprises a document.



Taj Mahal

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Coordinates: 27°10′30″N 78°2′31″E

For other uses, see Taj Mahal (disambiguation).

The Taj Mahal (/tɑːdʒ məˈhɑːl, tɑːʒ-/ TAHJ mə-HAHL, TAHZH -; Hindustani: [t̪aːdʒ ˈmɛɦ(ɛ)l]; lit. 'Crown of the Palace') is an ivory-white marble mausoleum on the right bank of the river Yamuna in Agra, Uttar Pradesh, India. It was commissioned in 1631 by the fifth Mughal emperor, Shah Jahan (r. 1628–1658), to house the tomb of his beloved wife, Mumtaz Mahal; it also houses the tomb of Shah Jahan himself. The tomb is the centrepiece of a 17-hectare (42-acre) complex, which includes a mosque and a guest house, and is set in formal gardens bounded on three sides by a crenellated wall.

Construction of the mausoleum was completed in 1648, but work continued on other phases of the project for another five years. The first ceremony held at the mausoleum was an observance by Shah Jahan, on 6 February 1643, of the 12th anniversary of the death of Mumtaz Mahal. The Taj Mahal complex is believed to have been completed in its entirety in 1653 at a cost estimated at the time to be around ₹32 million, which in 2015 would be approximately ₹52.8 billion (US$827 million).[4]

The building complex incorporates the design traditions of Indo-Islamic and Mughal architecture. It employs symmetrical constructions with the usage of various shapes and symbols. While the mausoleum is constructed of white marble inlaid with semi-precious stones, red sandstone was used for other buildings in the complex similar to the Mughal era buildings of the time. The construction project employed more than 20,000 workers and artisans under the guidance of a board of architects led by Ustad Ahmad Lahori, the emperor's court architect.

The Taj Mahal was designated as a UNESCO World Heritage Site in 1983 for being "the jewel of Islamic art in India and one of the universally admired masterpieces of the world's heritage". It is regarded as one of the best examples of Mughal architecture and a symbol of Indian history. The Taj Mahal is a major tourist attraction and attracts more than five million visitors a year. In 2007, it was declared a winner of the New 7 Wonders of the World initiative. The Taj Mahal and its setting, surrounding grounds, and structures are a Monument of National Importance, administered by the Archaeological Survey of India.[5]

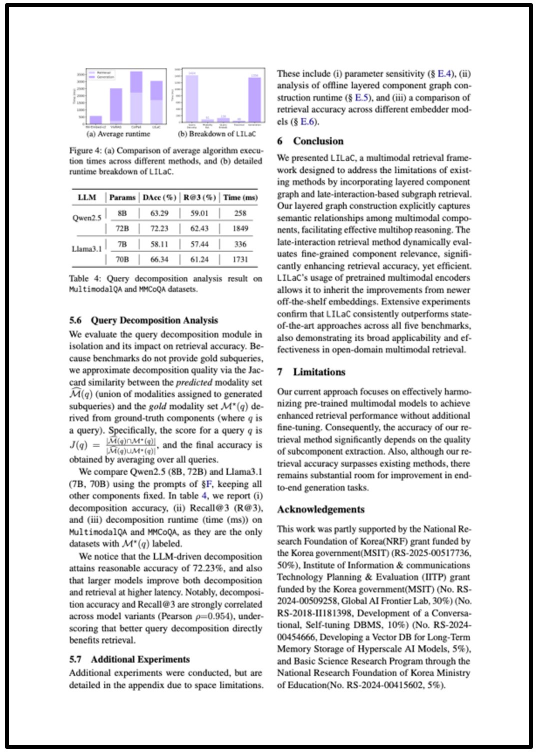
Taj Mahal

Location	Agra, Uttar Pradesh, India
Coordinates	27°10′30″N 78°2′31″E
Area	17 hectares (42 acres)[1]
Height	73 m (240 ft)
Built	1631–1653; 372 years ago[2]
Built for	Mumtaz Mahal
Architect	Ustad Ahmad Lahori
Architectural style	Mughal architecture
Visitors	4.84 million [3] (in Apr '22 - Feb '23)
Governing body	Ministry of Culture, Government of India (via Archaeological Survey of India)
Website	tajmahal.gov.in

Monument of National Importance

Official name	Taj Mahal and grounds (including the Masjid on the west side, the pavilions on the east and west sides of the grounds)
Reference no.	N-UP-A28

An example webpage



(a) Average runtime (b) Breakdown of LILaC

Figure 4: (a) Comparison of average algorithm execution times across different methods, and (b) detailed runtime breakdown of LILaC.

LLM	Params	DAcc (%)	R@3 (%)	Time (ms)
Qwen2.5	8B	63.29	59.01	258
	72B	72.23	62.43	1849
Llama3.1	7B	58.11	57.44	336
	70B	66.34	61.24	1731

Table 4: Query decomposition analysis result on MultimodalQA and MMCoQA datasets.

5.6 Query Decomposition Analysis

We evaluate the query decomposition module in isolation and its impact on retrieval accuracy. Because benchmarks do not provide gold subqueries, we approximate decomposition quality via the Jaccard similarity between the *predicted* modality set $\hat{\mathcal{M}}(q)$ (union of modalities assigned to generated subqueries) and the *gold* modality set $\mathcal{M}^*(q)$ derived from ground-truth components (where q is a query). Specifically, the score for a query q is $J(q) = \frac{|\hat{\mathcal{M}}(q) \cap \mathcal{M}^*(q)|}{|\hat{\mathcal{M}}(q) \cup \mathcal{M}^*(q)|}$, and the final accuracy is obtained by averaging over all queries.

We compare Qwen2.5 (8B, 72B) and Llama3.1 (7B, 70B) using the prompts of §F, keeping all other components fixed. In table 4, we report (i) decomposition accuracy, (ii) Recall@3 (R@3), and (iii) decomposition runtime (time (ms)) on MultimodalQA and MMCoQA, as they are the only datasets with $\mathcal{M}^*(q)$ labeled.

We notice that the LLM-driven decomposition attains reasonable accuracy of 72.23%, and also that larger models improve both decomposition and retrieval at higher latency. Notably, decomposition accuracy and Recall@3 are strongly correlated across model variants (Pearson ρ=0.954), underscoring that better query decomposition directly benefits retrieval.

5.7 Additional Experiments

Additional experiments were conducted, but are detailed in the appendix due to space limitations. These include (i) parameter sensitivity (§ E.4), (ii) analysis of offline layered component graph construction runtime (§ E.5), and (iii) a comparison of retrieval accuracy across different embedder models (§ E.6).

6 Conclusion

We presented LILaC, a multimodal retrieval framework designed to address the limitations of existing methods by incorporating layered component graph and late-interaction-based subgraph retrieval. Our layered graph construction explicitly captures semantic relationships among multimodal components, facilitating effective multihop reasoning. The late-interaction retrieval method dynamically evaluates fine-grained component relevance, significantly enhancing retrieval accuracy, yet efficient. LILaC's usage of pretrained multimodal encoders allows it to inherit the improvements from newer off-the-shelf embeddings. Extensive experiments confirm that LILaC consistently outperforms state-of-the-art approaches across all five benchmarks, also demonstrating its broad applicability and effectiveness in open-domain multimodal retrieval.

7 Limitations

Our current approach focuses on effectively harmonizing pre-trained multimodal models to achieve enhanced retrieval performance without additional fine-tuning. Consequently, the accuracy of our retrieval method significantly depends on the quality of subcomponent extraction. Also, although our retrieval accuracy surpasses existing methods, there remains substantial room for improvement in end-to-end generation tasks.

Acknowledgements

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An example PDF file

Multimodal Document Question Answering

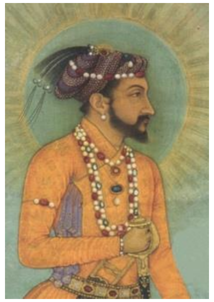
Correctly answering the given question using information of multimodal components

In the mausoleum built by the Mughal emperor who married Mumtaz Mahal, how many slender minarets surround the central dome?

Four

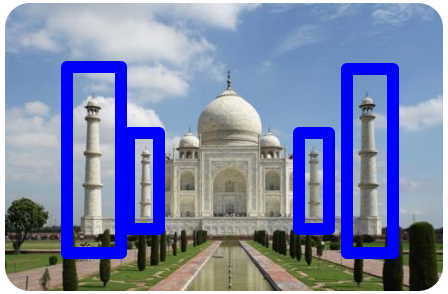
Multimodal Documents

Shah Jahan



Shah Jahan commissioned many monuments, including the Red For
and the Taj Mahal, where his favorite consort Mumtaz Mahal is entombed.

Taj Mahal



The Taj Mahal is an ivory white marble mausoleum on the right bank of the river Yamuna in Agra, India.

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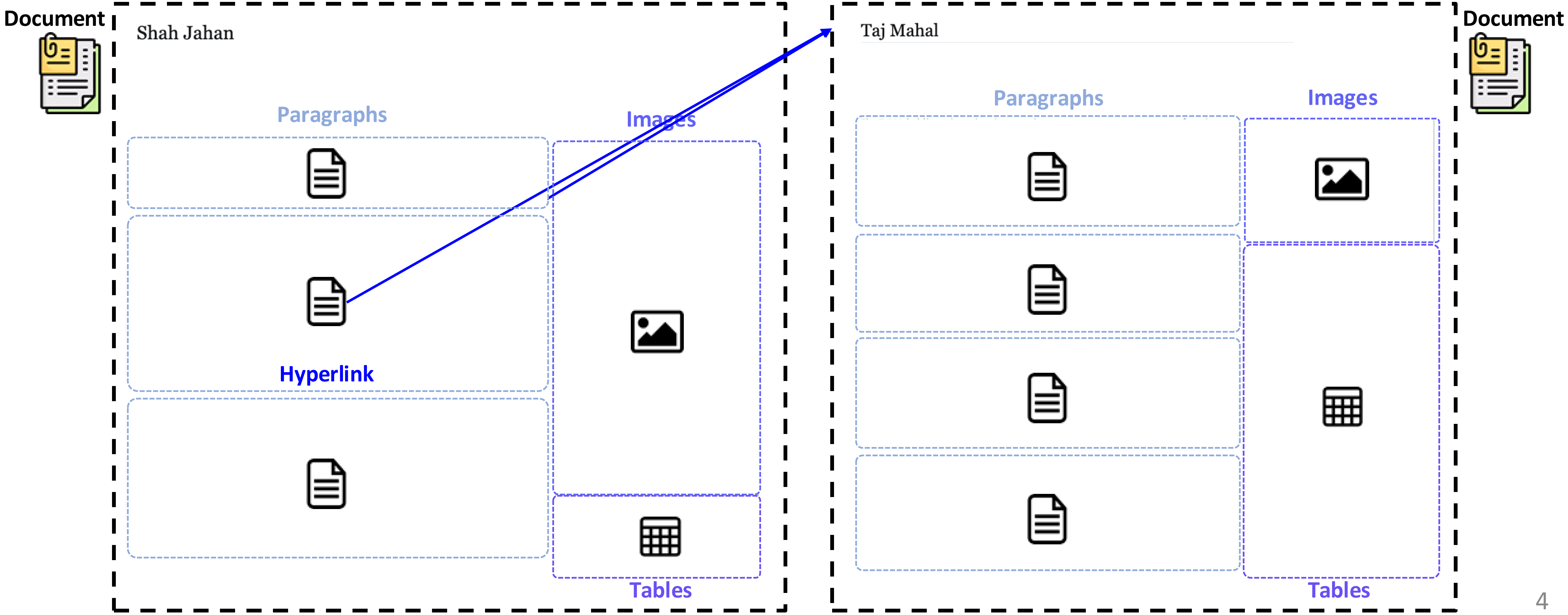
How Can We Model Multimodal Documents?

- Component: One of paragraph, table or an image
- Document: A set of components

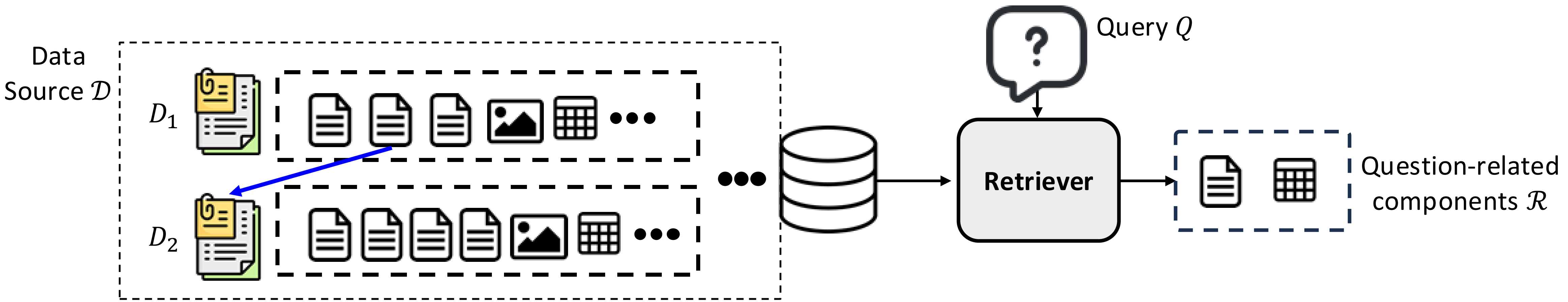


Our Task

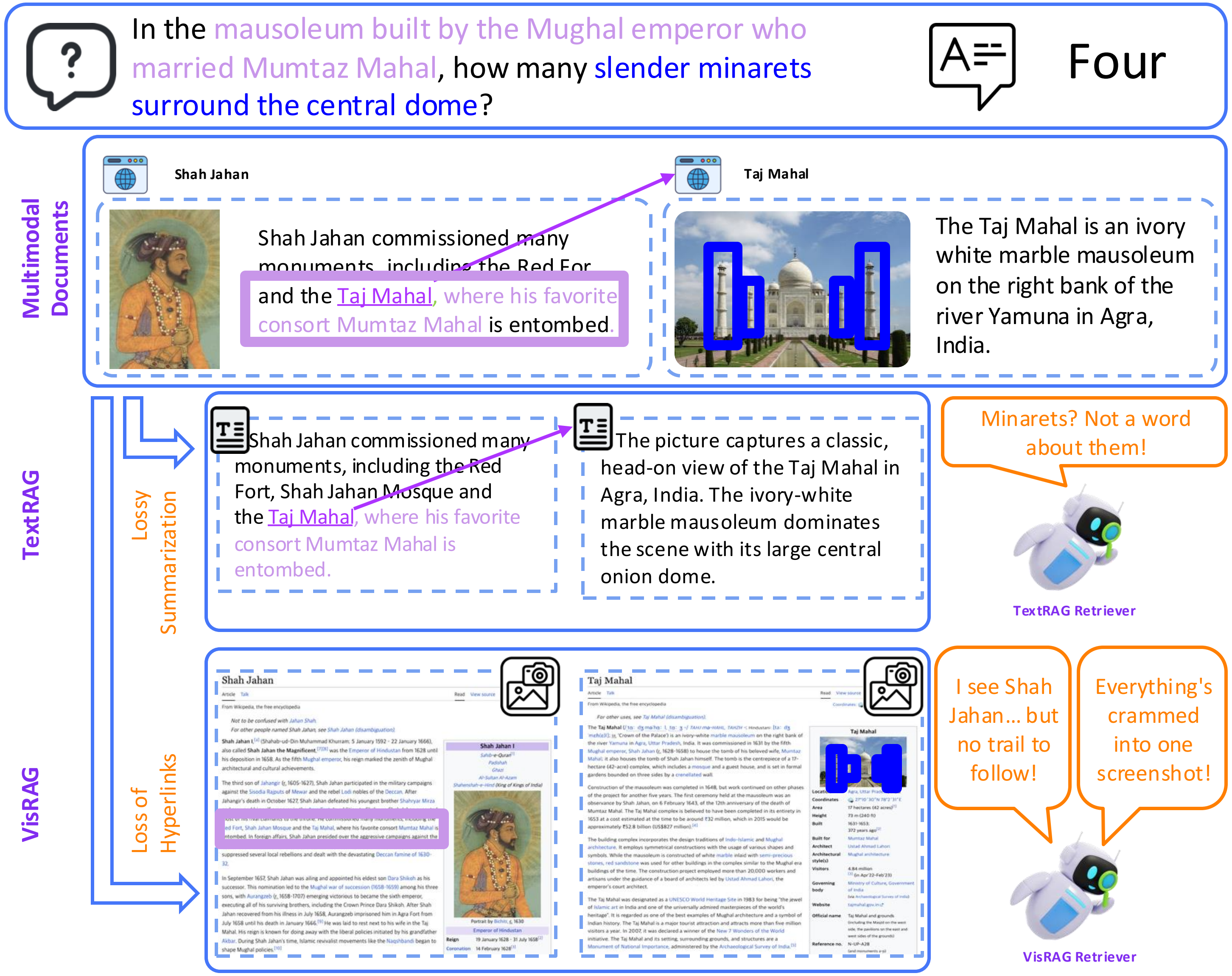
Open-Domain Multimodal Document Retrieval

Given a data source $\mathcal{D}$, a link mapping $\mathcal{L}$ and a query Q, correctly retrieve a ranked list of components $\mathcal{R} = [C_1, \dots, C_{n_{ret}}]$. $\mathcal{R}$ should contain the ground truth set of components $\{C_{gt_1}, \dots, C_{gt_r}\}$.

- Data source $\mathcal{D} = \{D_1, \dots, D_{k_{doc}}\}$
- Document $D = \{C_1, \dots, C_{k_{comp}}\}$
- Component $C \in \{P, T, I\}$
  - P: paragraph, T: table, I: image
- Link mapping $\mathcal{L} = \mathcal{C} \rightarrow \mathcal{D}$
  - $\mathcal{C}$: A set of all components within $\mathcal{D}$

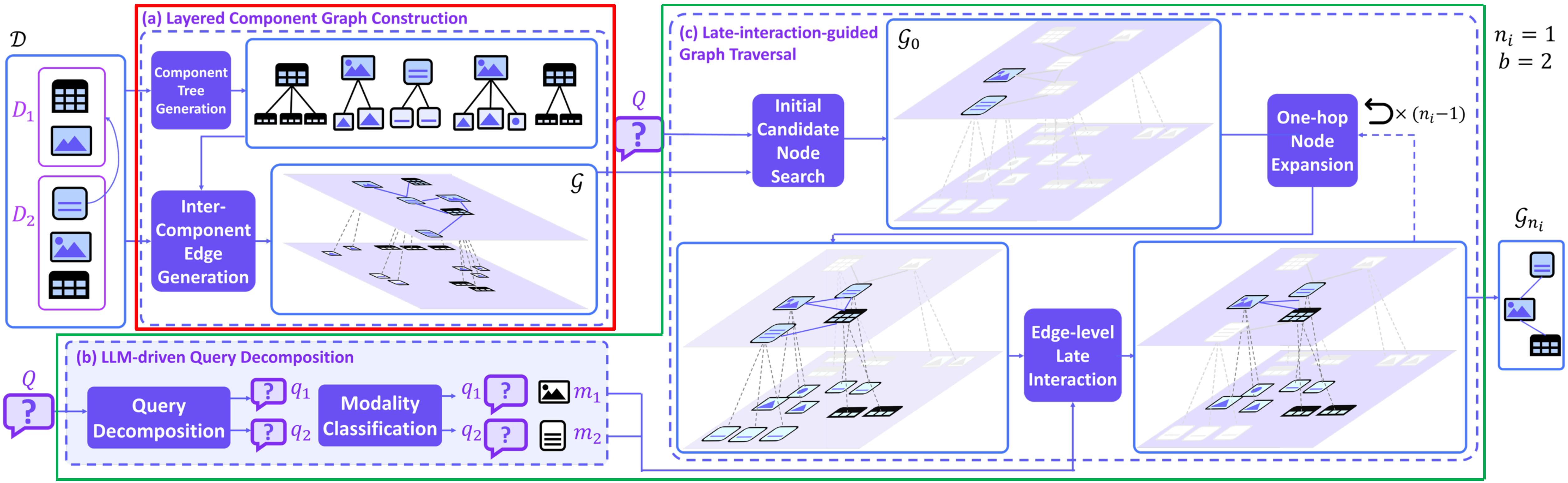


Problems of Existing Work



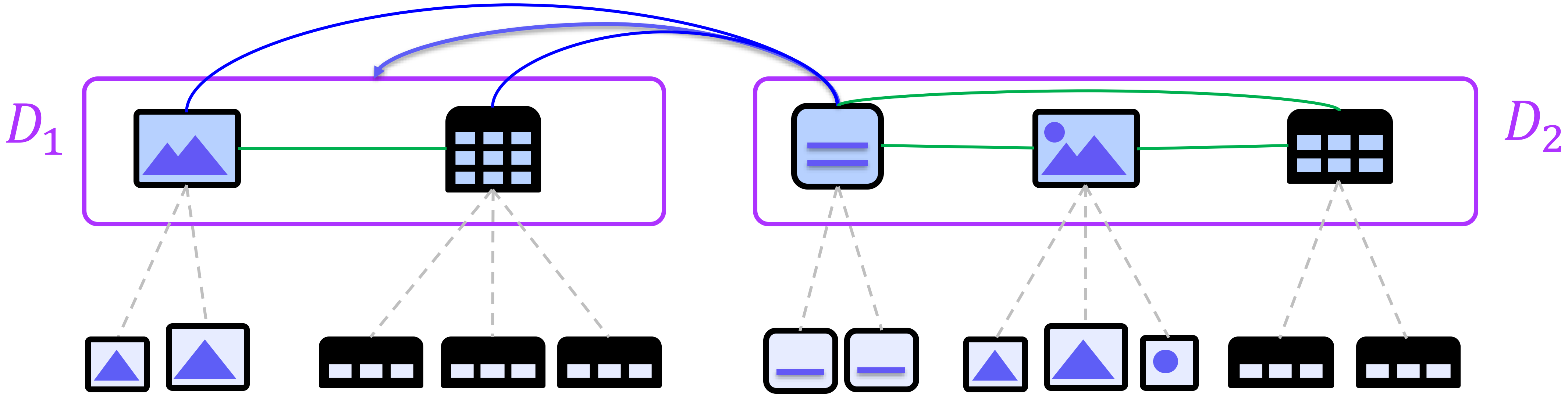
Overview of LILaC

- **Offline:** generate a graph data structure
- **Online**: retrieve a query-relevant subgraph from the graph



What Should We Consider for the Graph?

- **Modality preservation**
- **Multimodal multihop reasoning**: Exploit both inter-document links and intra-document relationships
- **Consideration of fine-grained granularity** [1]: Paragraph -> sentences, Table -> rows, Image -> detected objects



[1] Chen, Tong, et al. "Dense x retrieval: What retrieval granularity should we use?." EMNLP (2024).

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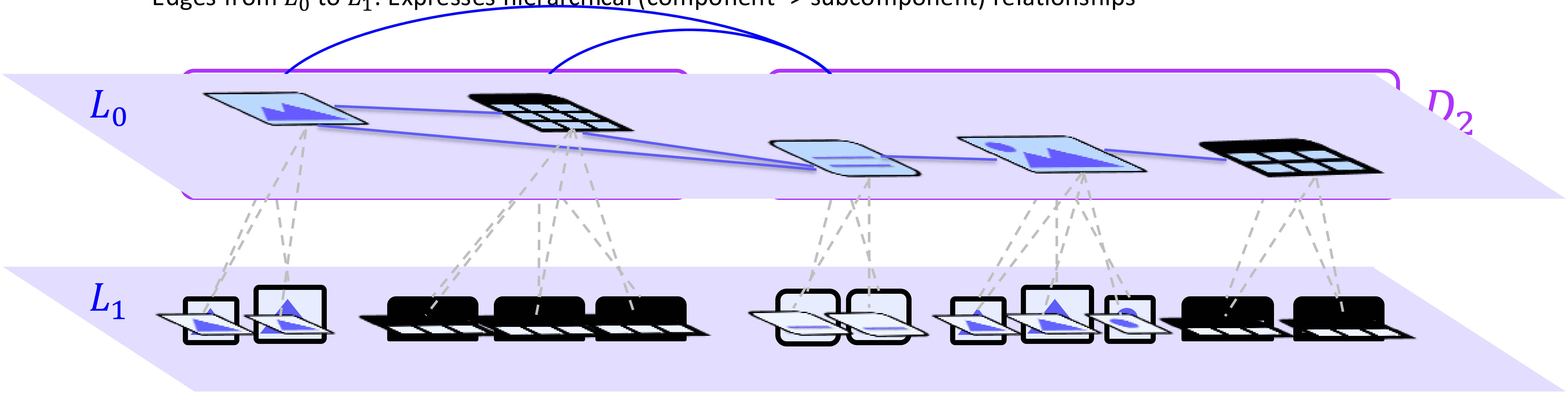
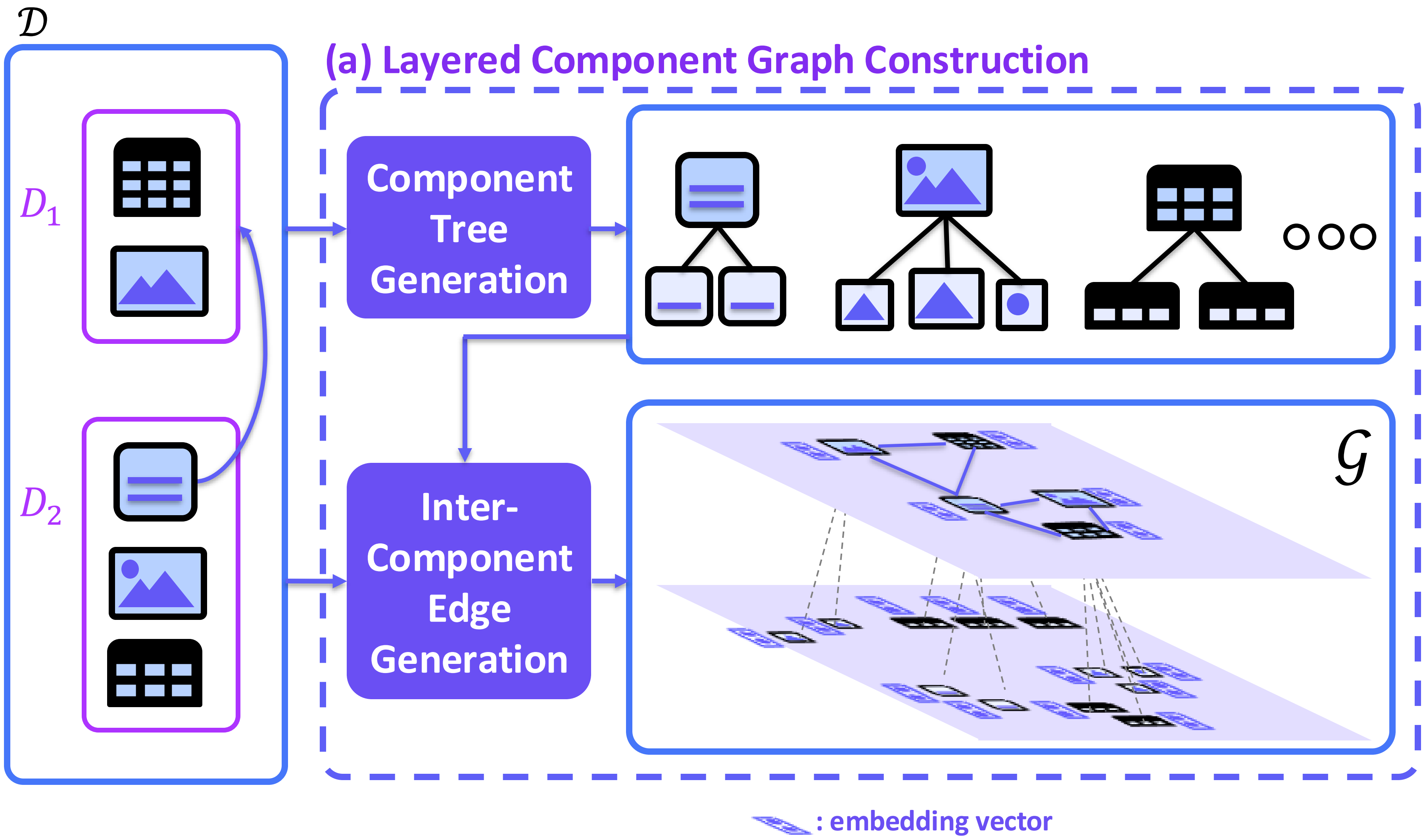
Key Data Structure

Layered Component Graph $\mathcal{G}$

- **Node:** component or subcomponent (e.g., sentence, row, image detection)
  - **Two layers**: each layer contains nodes for different granularity (coarse-grained L_0 vs fine-grained L_1)
- **Edges:**
  - Edges within L_0 :Expresses the hyperlink relationship and intra-document relationships
  - Edges from L_0 to L_1: Expresses hierarchical (component -> subcomponent) relationships



Layered Component Graph Construction



Component Tree Generation

1. Break down components into subcomponents
   - Passage -> Split into sentences
   - Table -> Split into rows
   - Image -> Detect subimages
2. Connect each component with its subcomponents.

Inter-Component Edge Generation

Generate an undirected edge for

- Every pair of components within the same document.
- Every pair of component with its linked document's components.

Embedding Generation

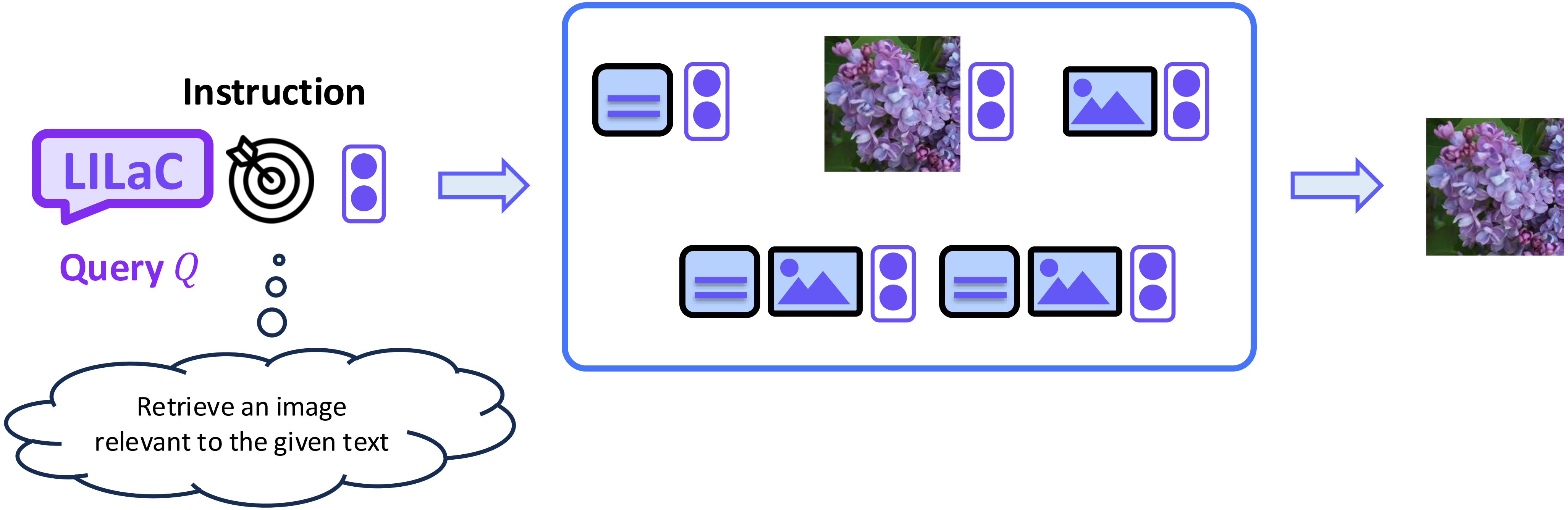
Background

Multimodal Embedders

They embed a *natural language sequence*, an *image*, or a *combination of two* into a single vector.

They require an instruction for each retrieval.

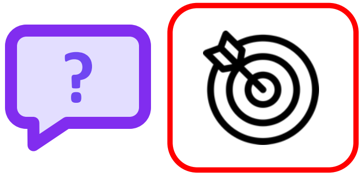
- Instruction includes information about the *target modality* of the retrieval.

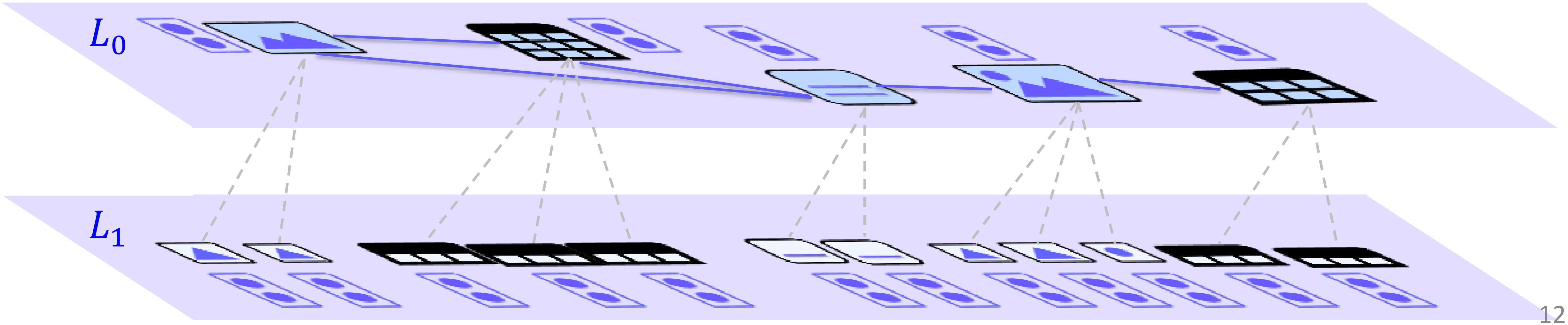


What Should We Consider for the Retrieval Algorithm?

- How should we find the instructions for multimodal embedding?
- ➢ **Challenge 1**: The instructions are not given in our setting
- How should we find subgraphs that need multimodal multihop reasoning?
- ➢ **Challenge 2**: The target subgraph may consist of multiple multimodal nodes (multihop reasoning)

Query Q

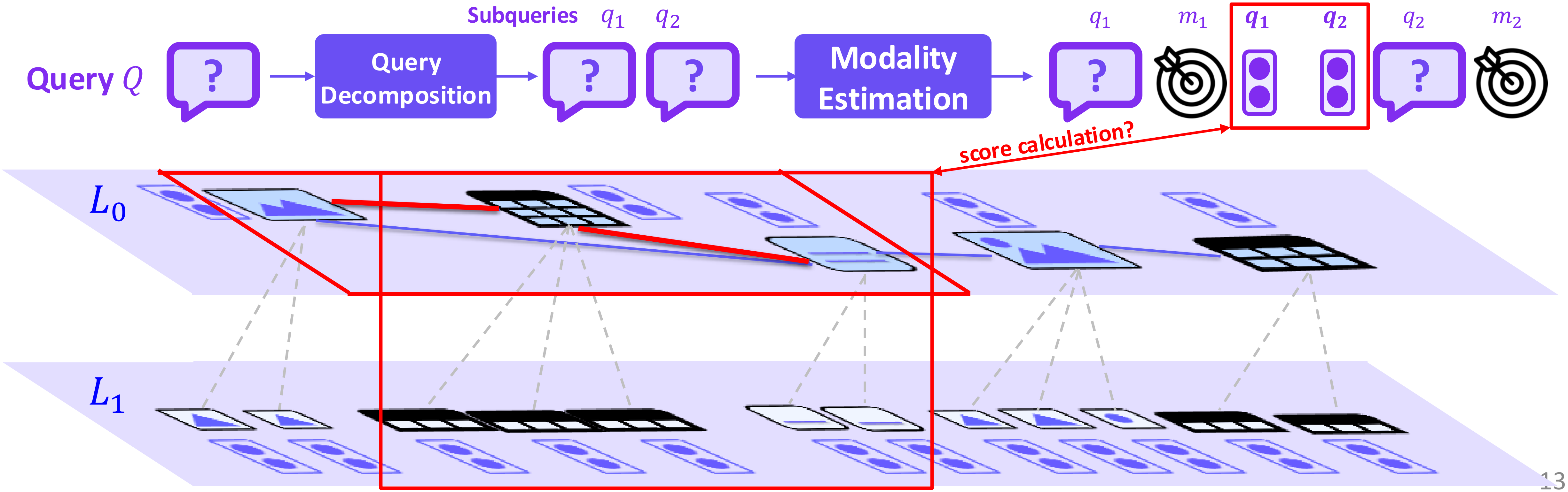




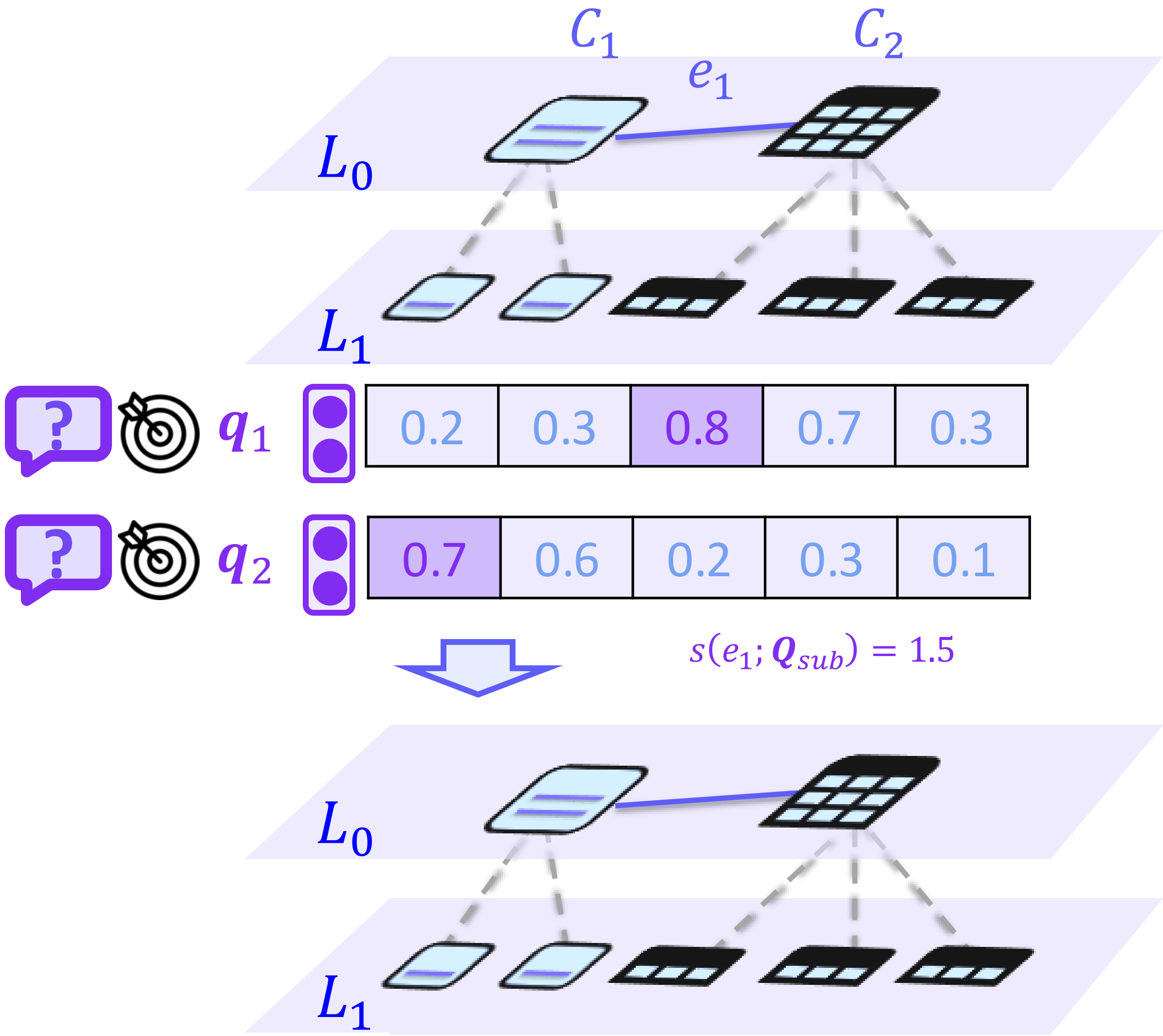
Key Idea

LILaC Retrieval Algorithm

- **Query decomposition & modality estimation**: extract subqueries, then perform modality estimation for each subquery.
- **Edge-based subgraph retrieval**: retrieves query-relevant edges, then merges them into a subgraph



Edge-level Late Interaction



Goal: Similarity calculation of a query and an edge

Input:

- Subquery embeddings $Q_{sub} = \{q_1, q_2, \ldots, q_n\}$
- Edge $e = (C_a, C_b)$
  - Subcomponent embeddings $e_c = \{c_{a,1}, \ldots, c_{a,k}\} \cup \{c_{b,1}, \ldots, c_{b,l}\}$

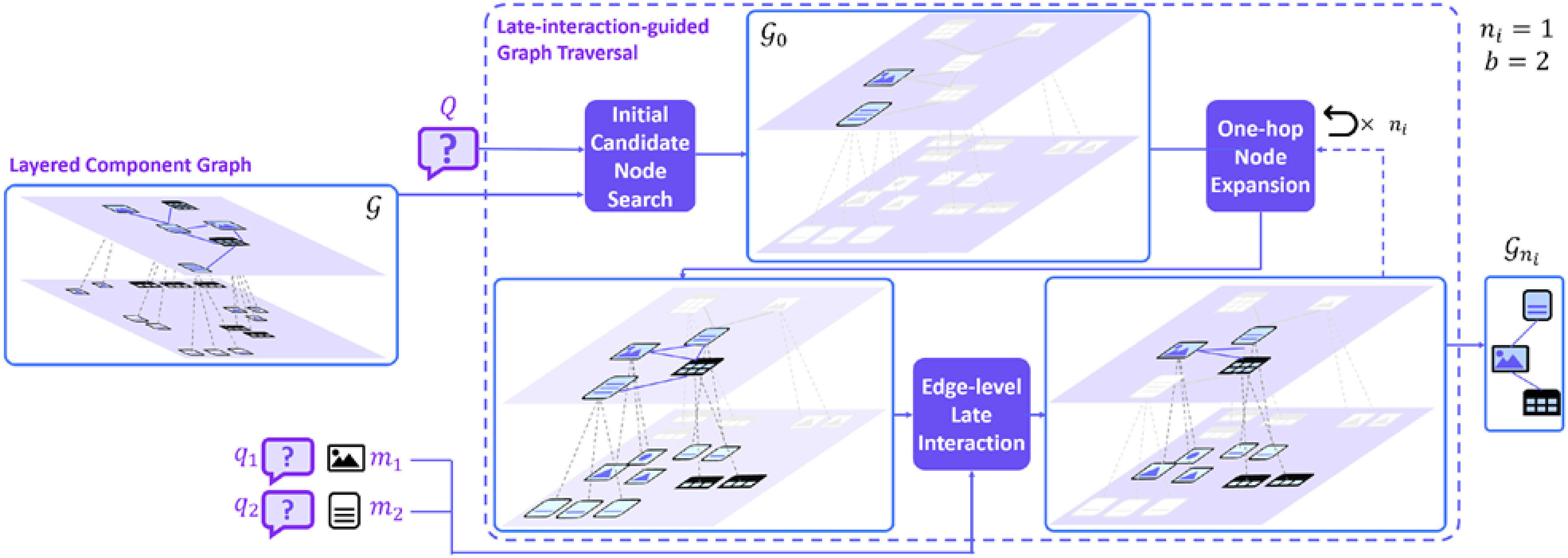
Motivation: A subquery is likely to show the highest similarity with its corresponding golden subcomponent.

Late-interaction-based score calculation:

$$s(e; Q_{sub}) = \sum_{q \in Q_{sub}} \max_{c \in e_c} sim(c, q)$$

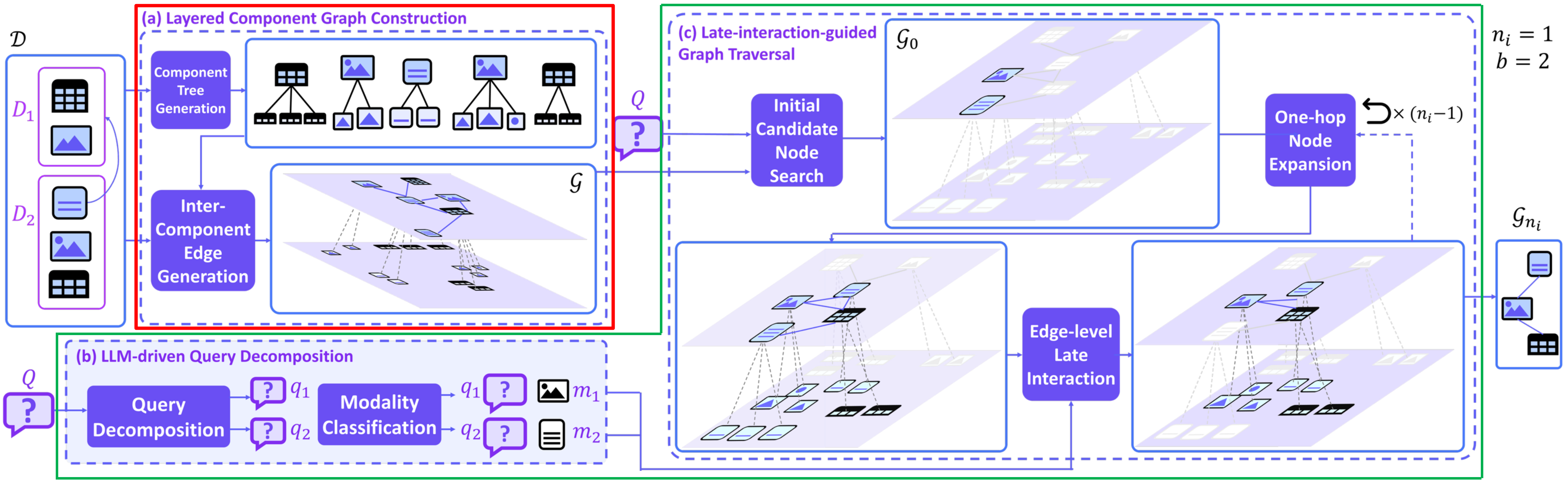
Late-Interaction-Guided Graph Traversal

1. Initiate the subgraph by performing *kNN* search over components
2. Iteratively
   a. Expand the subgraph by performing a one-hop node expansion from existing nodes.
   b. Leave top-b query-relevant edges among the expanded graph.



Overview of LILaC (Revisited)

- **Offline:** generate a graph data structure
- **Online**: retrieve a query-relevant subgraph from the graph



Retriever Accuracy Comparison

- LILaC achieves SOTA performance on all 5 benchmarks.
- LILaC outperforms the previous VisRAG SOTA models, VisRAG-Ret and ColPali, by substantial margins of 12.39% and 9.85% in R@3, and 14.45% and 10.49% in MRR@10, on average, respectively.

Algorithm	Embedder Type	MP-DocVQA		SlideVQA		InfoVQA		MultimodalQA		MMCoQA	
		R@3	MRR@10	R@3	MRR@10	R@3	MRR@10	R@3	MRR@10	R@3	MRR@10
NV-Embed-v2	Text	67.85	61.91	88.49	79.55	86.21	80.86	60.19	67.86	46.16	41.45
VisRAG-Ret	Image	83.25	75.55	91.55	84.30	92.76	86.22	50.08	55.08	27.63	23.75
ColPali		80.71	74.86	89.39	81.55	88.30	82.76	58.73	65.05	36.24	32.33
LILaC (w/ mmE5)	Multimodal	61.25	55.30	77.52	68.80	75.09	69.86	54.79	59.02	48.88	40.30
LILaC (w/ UniME)		77.83	71.42	84.35	77.93	82.28	75.27	58.52	61.44	49.63	42.97
LILaC (w/ MM-Embed)		**83.59**	**78.75**	**92.81**	**84.43**	**93.17**	**86.83**	**69.07**	**75.28**	**55.80**	**50.77**

Table 1: Retrieval accuracy (Recall@3 (*R@3*) and *MRR@10*) of LILaC and its competitors on five benchmarks. The best score in each column is in **bold**. The in-domain fine-tuned settings are colored in orange.

End-to-End QA Accuracy Comparison

- LILaC achieves SOTA average end-to-end accuracy, with average EM and F1 score 9, respectively.
- 17.40% and 18.47% accuracy raise compared to the previously best-performi VL

More interesting experiments in our paper!

Algorithm	MLLM	MP-DocVQA		SlideVQA		InfoVQA		MultimodalQA		MMCoQA	
		EM	F1	EM	F1	EM	F1	EM	F1	EM	F1
NV-Embed-v2	Qwen2.5-VL 7B	56.51	63.16	53.77	64.41	60.72	63.40	37.23	43.85	28.05	34.67
VisRAG-Ret	MiniCPM V2.6	54.31	68.86	43.88	62.37	50.83	57.55	28.18	34.01	21.51	27.87
VisRAG-Ret	Qwen2.5-VL 7B	65.34	72.24	55.03	66.13	60.16	61.93	22.24	25.55	16.69	20.90
ColPali	Qwen2.5-VL 7B	64.46	71.16	53.77	64.54	58.07	60.38	23.59	27.37	18.07	22.30
LILaC (w/ mmE5)	Qwen2.5-VL 7B	52.96	59.53	50.89	59.07	50.12	53.12	40.72	47.46	33.90	40.38
LILaC (w/ UniME)	Qwen2.5-VL 7B	62.43	69.40	53.05	62.89	53.39	56.86	43.42	49.72	33.39	40.12
LILaC (w/ MM-Embed)	Qwen2.5-VL 7B	**65.48**	**72.42**	**55.57**	**66.32**	**60.91**	**62.87**	**44.57**	**51.97**	**36.31**	**43.22**

Table 2: End-to-end accuracy (EM and F1) of LILaC and its competitors for the 5 benchmarks. The best score in each column is in **bold**. Generation results corresponding to in-domain fine-tuned settings are colored in orange.

Thank You